04. Preprocessing & Feature Engineering

Understand every transformation and the order in which it was applied.

Project: CreditWise – Loan Approval Prediction

Source: supplied Loan-Approval-main.zip | Notebook: credit_wise.ipynb | Dataset: loan_approval_data.csv

Missing-value handling in notebook

The notebook identifies numeric and object columns. SimpleImputer(strategy='mean') fills numeric missing values, and SimpleImputer(strategy='most_frequent') fills object missing values. This produces a complete table of 1,000 rows. However, because the target Loan_Approved is object dtype, its 50 missing values are also filled with the most frequent target class. This is a critical methodological error.

Correct treatment

- Separate rows with known target from rows with unknown target before supervised modelling.
- Never impute a missing target label as though it were observed; investigate label collection and optionally use unlabelled rows for separate semi-supervised approaches only with a defensible design.
- Fit imputers on training data only and apply the learned transformations to validation/test data.
- Consider median imputation for skewed numeric fields and explicit missingness indicators where missingness may be informative.
- For categorical fields, use most-frequent or an explicit Unknown category, based on the missingness mechanism.

Identifiers and encoding

Applicant_ID is dropped. Education_Level and Loan_Approved are label-encoded. The target encoding is fine for a binary target if mapping is recorded. Education is nominal in this dataset; label encoding imposes an arbitrary order, so one-hot encoding is usually safer unless a genuine ordinal hierarchy is defined. Employment_Status, Marital_Status, Loan_Purpose, Property_Area, Gender and Employer_Category are one-hot encoded with drop='first' and handle_unknown='ignore'.

Scaling

StandardScaler is fitted on X_train and applied to X_test, which is the correct fit/transform pattern for the split that exists. Standardization is important for Logistic Regression and KNN because feature units vary widely. GaussianNB can also be trained on scaled values, though scaling is not generally required for Gaussian Naive Bayes.

Feature engineering iteration

The notebook creates DTI_Ratio_sq = DTI_Ratio² and Credit_Score_sq = Credit_Score². It then drops the original DTI_Ratio and Credit_Score columns from X, leaving only their squared versions. Squared terms can model curvature, but removing the original linear terms constrains the model in an unusual way. A better polynomial design usually retains both x and x², and validates whether the extra complexity improves out-of-sample performance.

Pipeline recommendation

Use a ColumnTransformer with numeric and categorical pipelines, fit only on the training fold, and combine it with the estimator in an sklearn Pipeline. This prevents preprocessing leakage during cross-validation and makes the full process reproducible.